

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2021

Bachelor in Computer Applications

Course Title: **Probability & Statistics**

Code No: Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

11. Discuss the role of statistics in computer application.
12. The following table shows the monthly salary of employees in a certain locality of Lalitpur Sub-metropolitan City:
13. Define the term correlation. Compute Karl Pearson's coefficient of correlation between advertisement cost and sales as per the data given below:
14. The following table gives the age of the computers of a certain company and annual maintenance costs:
15. Fit a Poisson distribution of the following data and calculate the expected frequencies:
16. Determine First Quartile (Q_1), D_7 and P_{80} (80th percentile) from the following data:
17. A box contains 30 items of which 10 are defectives. If two items are selected randomly from the box without replacement, what is the probability that (a) Both are defective (b) Both are non-defective

Group C

Attempt any TWO questions [2x10=20]

18. The following data represents the scores made in an intelligence test by the groups of students from section A and section B of National College:
19. The lifetime of a certain electronic component is a normal random variate with an expectation of 5000 hours and a standard deviation of 100 hours. Compute the probabilities under the following conditions: (a) Lifetime of components is less than 3012 hours (b) Lifetime of components between 4000 to 6000 hours (c) Lifetime of components less than 4500 hours (d) Lifetime of components more than 7000 hours
20. Perform the ANOVA from Two-Way classification: